SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

Engineer to Excel

CSA0823-Python Programming (Slot C)

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Python Data Type, Variable, Function Program Questions

1. Swap without a third variable

Two variables a = 10 and b = 25 are given. Swap their values without using a temporary variable. Then explain why your method works.

Logic

- b's value (25) goes into a
- a's value (10) goes into b
- Both values are exchanged **at the same time**.

Sample Input

a: 10

b: 25

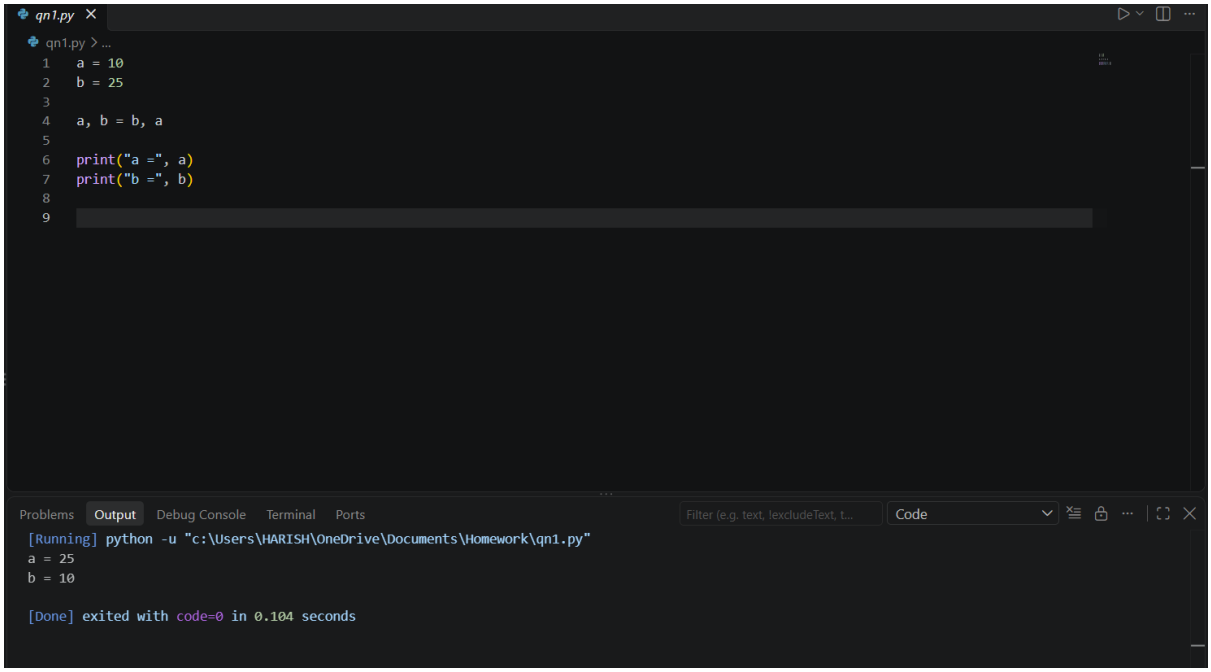
Sample Output

After swapping:

a = 25

b = 10

Result



2. Predict the type

Without running the code, write the data type of each result, then verify:

python

x = 5 / 2

y = 5// 2

z = "5" * 3

w = (5)

v = (5,)

Which ones surprised you, and why?

Logic

- 5 / 2 → Normal division gives a **decimal value** → float
- 5 // 2 → Floor division gives only the **whole-number part** → int
- "5" * 3 → Repeats the string 3 times → str

- (5) → Parentheses alone do **not** make a tuple → int
- (5,) → Comma makes it a **tuple** → tuple

Sample Input

number: 5

divisor: 2

string: 5

Sample Output

2.5 <class 'float'>

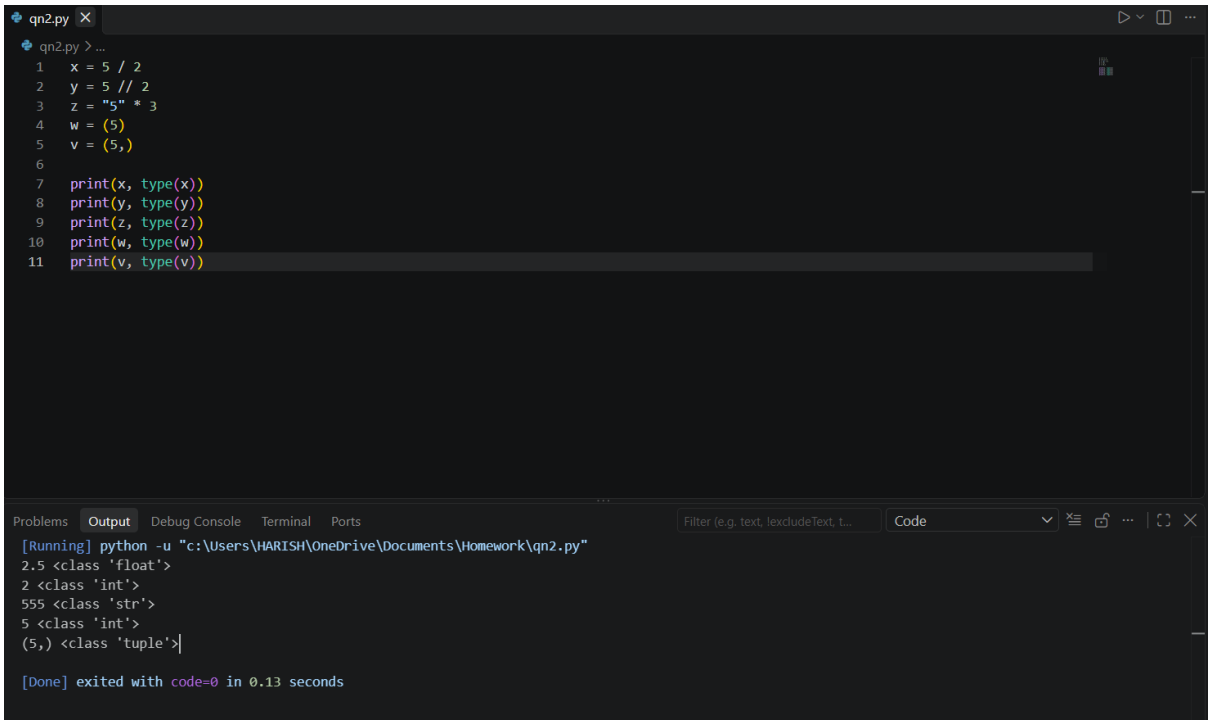
2 <class 'int'>

555 <class 'str'>

5 <class 'int'>

(5,) <class 'tuple'>

Result



3. The mutable trap

python

a = [1, 2, 3]

b = a

b.append(4)

print(a)

Predict the output. Then change the code so that a stays unchanged. Explain the difference between assignment and copying.

Logic

- a contains [1, 2, 3].
- b = a.copy() creates a **separate copy** of a.
- b.append(4) adds 4 only to b.
- Therefore, a remains unchanged.

Sample Input

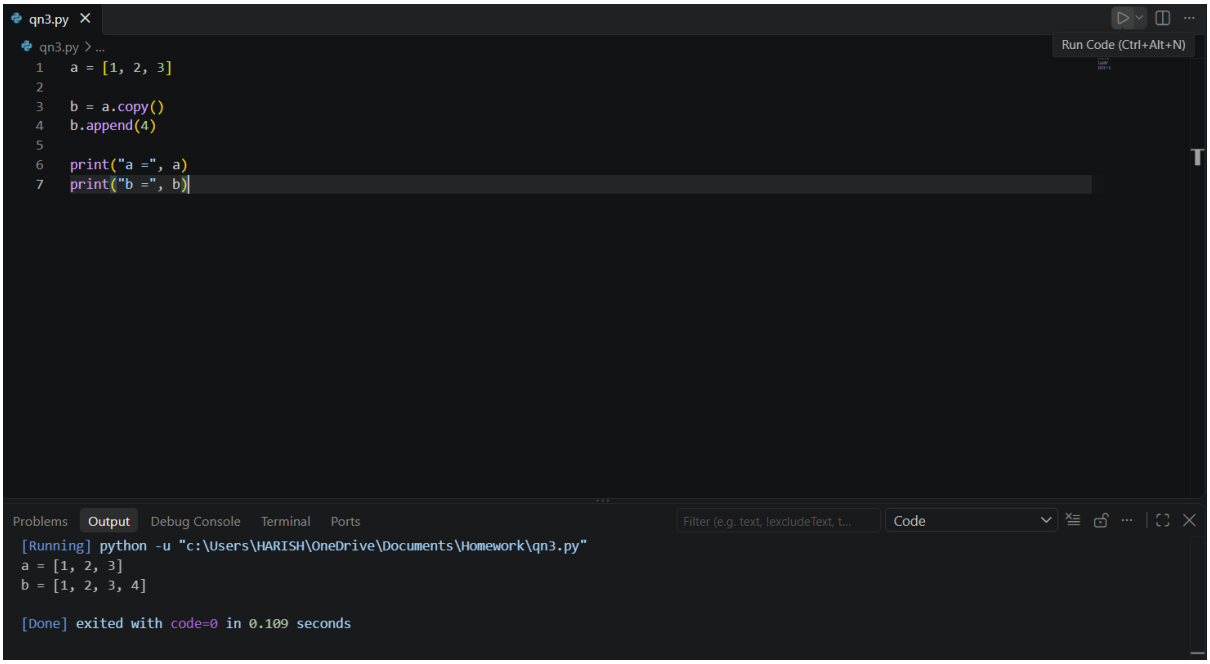
a = [1, 2, 3]

Sample Output

a = [1, 2, 3]

b = [1, 2, 3, 4]

Result



4. Temperature converter

Write a function convert(temp, unit) that converts Celsius to Fahrenheit if unit is "C", and Fahrenheit to Celsius if unit is "F". What should it do if the unit is anything else? Decide and justify.

Logic

- temp stores the temperature value.
- unit stores whether the input is "C" or "F".
- If unit is "C", convert Celsius → Fahrenheit.
- If unit is "F", convert Fahrenheit → Celsius.
- The output shows the converted **value + unit**.
- If the unit is anything else, display **"Invalid unit"**.

Sample Input

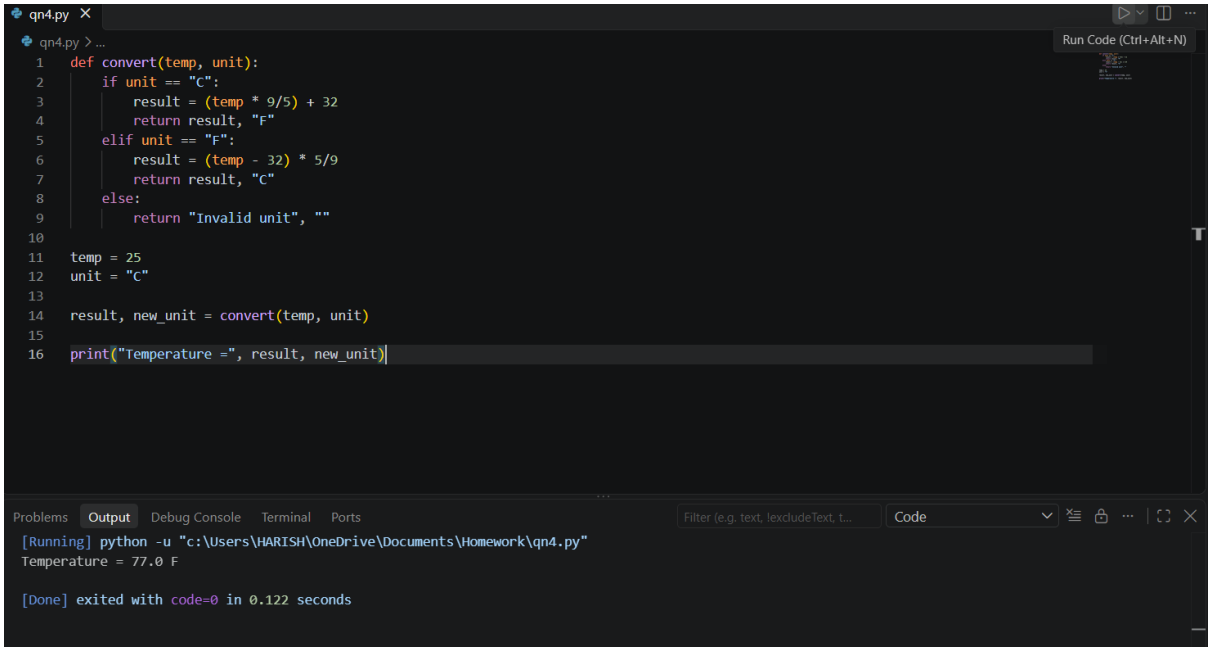
Enter temperature: 25

Enter unit (C/F): C

Sample Output

Temperature = 77.0 F

Result



5. String detective

Take any sentence typed by the user. Write a program (using variables and string methods) that prints the number of words, the longest word, and the sentence in reverse word order.

Logic

- split() → separates the sentence into words.
- len() → counts the number of words.
- max(..., key=len) → finds the longest word.
- [::-1] → reverses the order of the words.
- " ".join() → joins the reversed words back into a sentence.

Sample Input

Enter a sentence: PYTHON IS A VERY EASY PROGRAMMING LANGUAGE

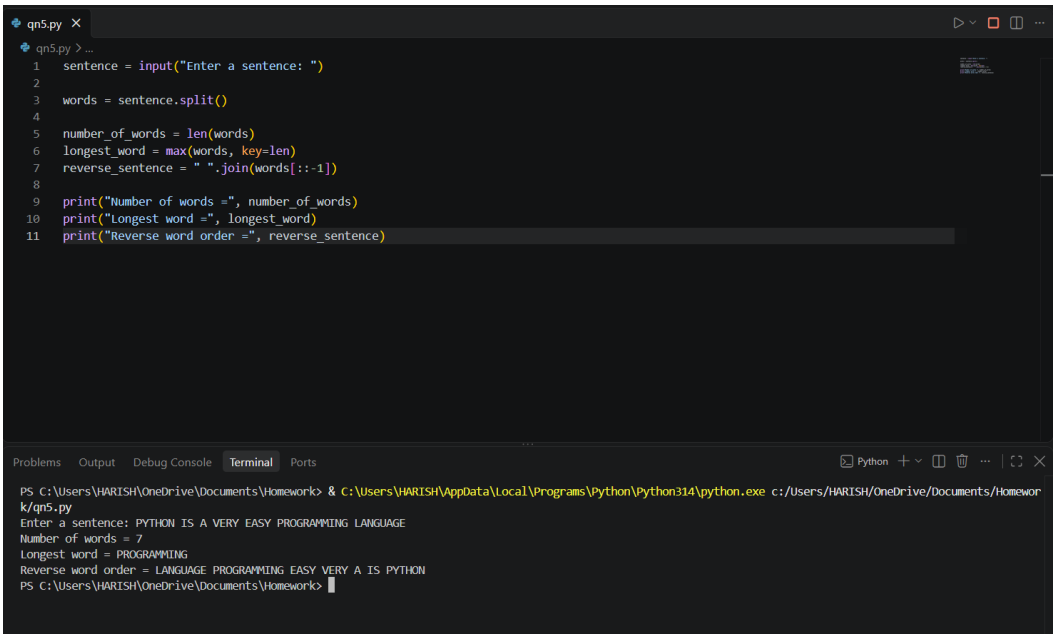
Sample Output

Number of words = 7

Longest word = PROGRAMMING

Reverse word order = LANGUAGE PROGRAMMING EASY VERY A IS PYTHON

Result



6. Fix the scope bug

python

```
count = 0
def increase():
    count = count + 1
increase()
print(count)
```

This code gives an error. Find out why, then fix it in **two different ways**. Which way is better practice?

Logic

Method 1 – global:

- count = 0 is outside the function.
- global count allows the function to change the outside count.
- count = count + 1 increases it to 1.

Method 2 – return:

- Send count into the function.
- Increase it by 1.
- Return the new value.
- Store the returned value back in count.

Sample Input

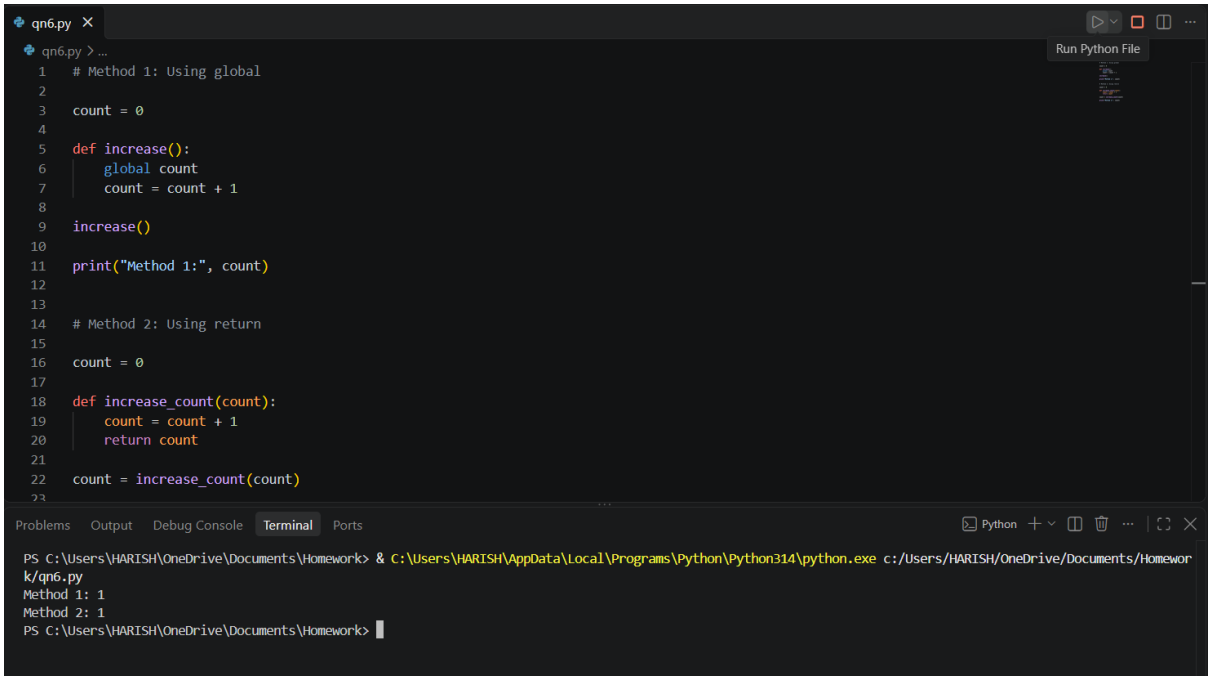
Initial count = 0

Sample Output

Method 1: 1

Method 2: 1

Result


qn6.py
Run Python File
Method 1: Using global
count = 0
def increase():
global count
count = count + 1
increase()
print("Method 1:", count)
Method 2: Using return
count = 0
def increase_count(count):
count = count + 1
return count
count = increase_count(count)
Problems
Output
Debug Console
Terminal
Ports
Python
PS C:\Users\HARISH\OneDrive\Documents\Homework> & C:\Users\HARISH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/HARISH/OneDrive/Documents/Homework/qn6.py
Method 1: 1
Method 2: 1
PS C:\Users\HARISH\OneDrive\Documents\Homework>

7. Function with a default surprise
Write a function greet(name, greeting="Hello") and call it in four different ways (positional, keyword, using the default, and swapping argument order with keywords). Then write one call that would cause an error and explain why.

Logic

- greet(name, greeting="Hello") → name is required, while greeting has a default value of "Hello".
- **Positional:** Values are given according to their position.
- **Keyword:** Values are given using parameter names.
- **Default:** If greeting is not given, "Hello" is automatically used.
- **Swapping with keywords:** Keyword arguments can be written in any order.
- **Error:** Giving the same parameter more than once causes an error.

Sample Input

No input required.

Sample Output

Hi Harish

Good Morning Harish

Hello Harish

Welcome Harish

Traceback (most recent call last):

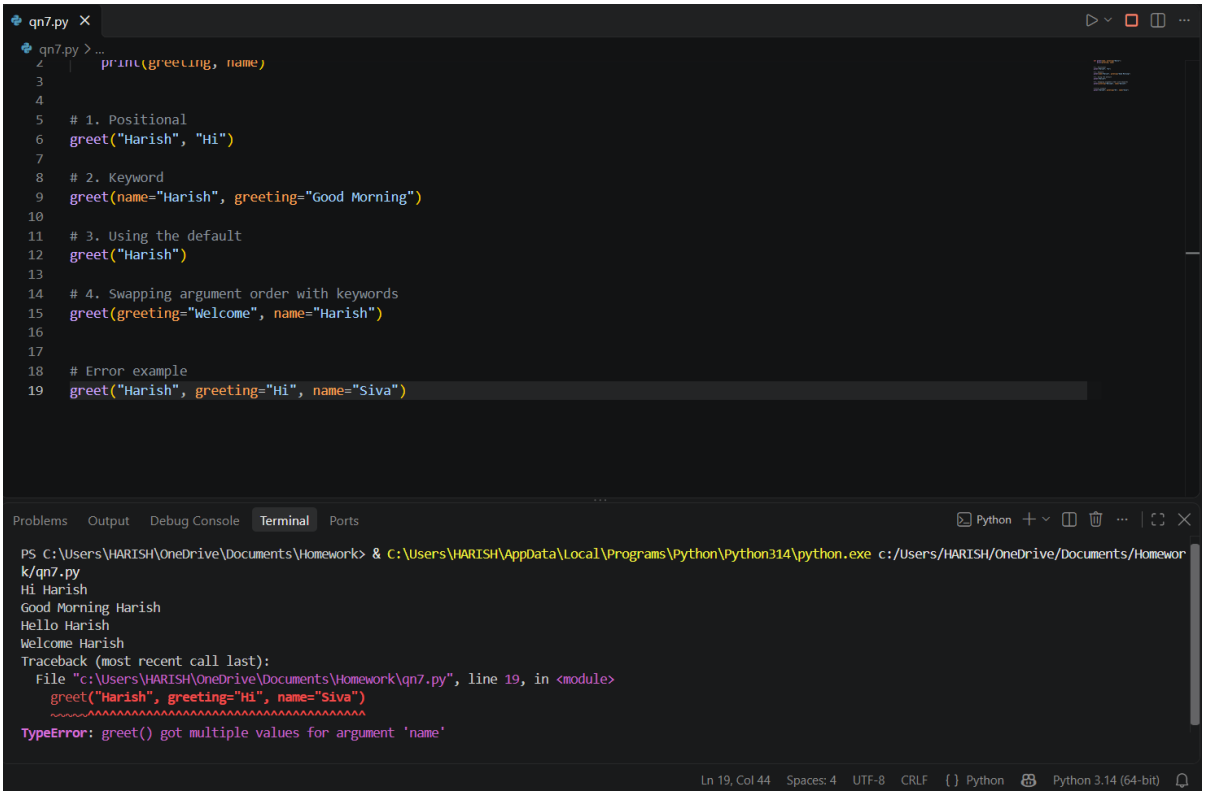
File "c:\Users\HARISH\OneDrive\Documents\Homework\qn7.py", line 19, in <module>

greet("Harish", greeting="Hi", name="Siva")

~~~~~^^

TypeError: greet() got multiple values for argument 'name'
~~~~~

Result



8. Data type chooser

You must store each of the following in a Python program. Choose the best data type (int, float, str, list, tuple, set, dict, bool) and justify your choice:

- a student's roll number
- marks in 5 subjects
- the days of the week
- unique email IDs of club members
- a student's name along with their age and class
- whether a library book is issued

Logic

- int(input()) → takes the **roll number** as an integer.
- input().split() → separates the **5 marks** entered with spaces.
- map(float, ...) → converts each mark into a decimal number.
- list(...) → stores all 5 marks in a **list**.
- tuple(...) → stores the days as a **tuple**.
- set(...) → stores email IDs and automatically removes duplicates.
- dict → stores student's **name, age, and class** using keys.
- == "True" → converts the book-issued answer into a **Boolean** (True or False).

Sample Input

Enter roll number: 51

Enter mark: 10

Enter mark: 20

Enter mark: 30

Enter mark: 40

Enter mark: 50

Enter days of the week: mon tue wd thurs fri sat sun

Enter email IDs separated by spaces: 123@simats.com 124@simats.com

Enter student name: harish

Enter age: 19

Enter class: ECE

Is the book issued? (True/False): true

Sample Output

--- Student Details ---

Roll number = 51

Marks = [10.0, 20.0, 30.0, 40.0, 50.0]

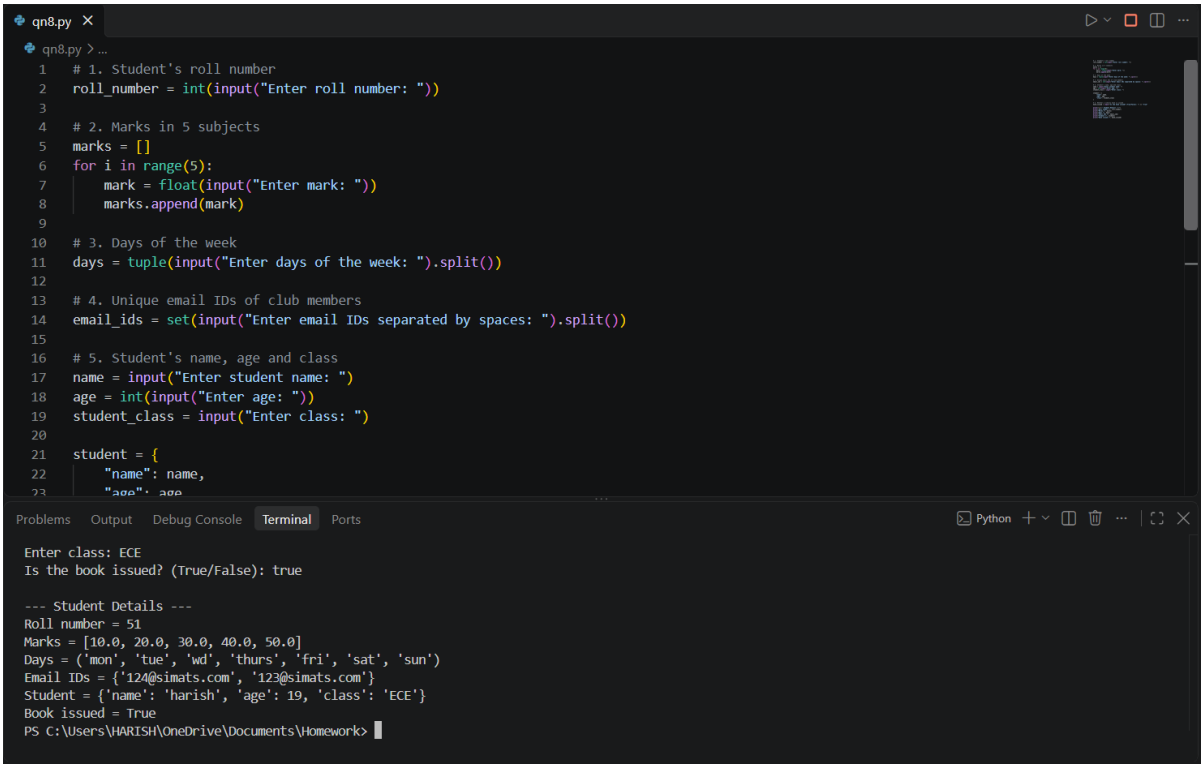
Days = ('mon', 'tue', 'wd', 'thurs', 'fri', 'sat', 'sun')

Email IDs = {'124@simats.com', '123@simats.com'}

Student = {'name': 'harish', 'age': 19, 'class': 'ECE'}

Book issued = True

Result



9. Return multiple values

Write a function analyze(numbers) that takes a list of numbers and returns the minimum, maximum, and average together. Show two different ways to receive and print the returned values. What data type does the function actually return?

Logic

- min(numbers) → finds the smallest number.
- max(numbers) → finds the largest number.
- sum(numbers) / len(numbers) → calculates the average.
- return minimum, maximum, average → returns all three values together.
- Python automatically stores the returned values as a **tuple**.

- **Method 1:** Unpack the tuple into three separate variables.
- **Method 2:** Store the complete tuple in one variable.
- type(result) → confirms that the returned data type is **tuple**.

Sample Input

Enter numbers: 10 20 30 40 50

Sample Output

Minimum = 10

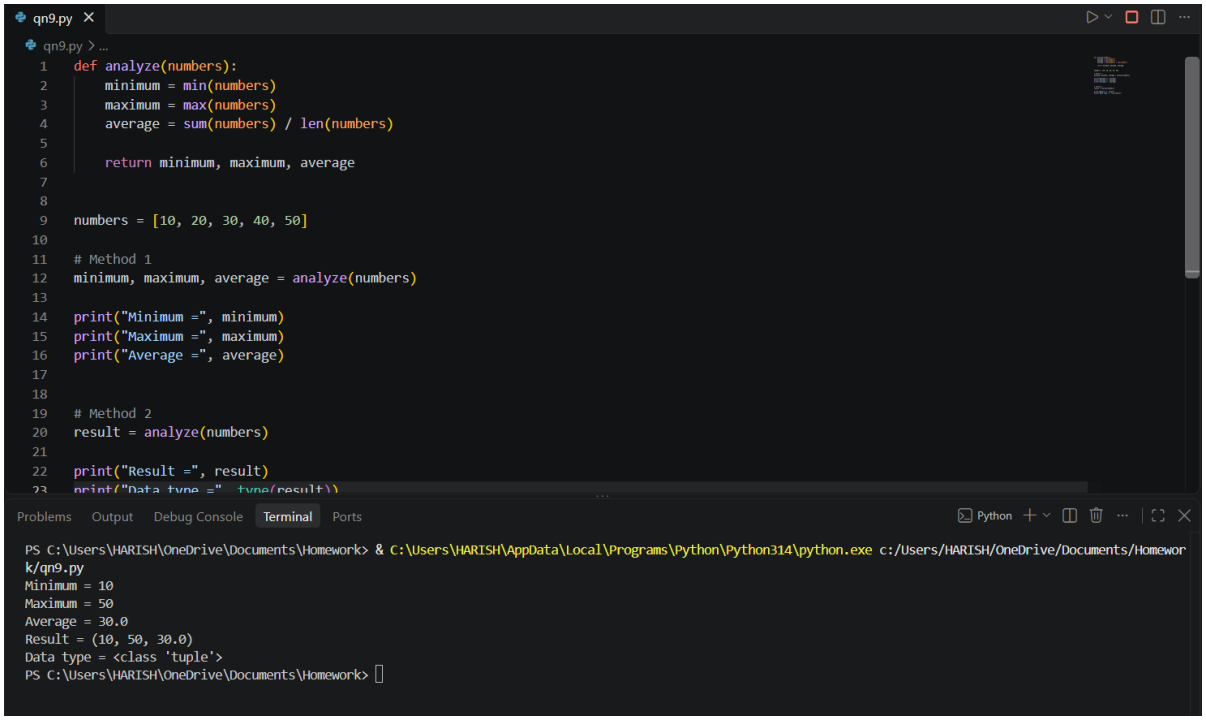
Maximum = 50

Average = 30.0

Result = (10, 50, 30.0)

Data type = <class 'tuple'>

Result



10. Build a mini-calculator with validation
Write separate functions for add, subtract, multiply, and divide. Ask the user for two numbers and an operation, and call the right function. Make sure the program doesn't crash on division by zero or when the user types letters instead of numbers. Hint: think about type conversion and where it can fail.

Logic

- Create separate functions for **add, subtract, multiply, and divide**.
- Get two numbers from the user using float().
- Get the operation (+, -, *, /).
- Use if-elif to select the correct function.
- For division, first check if the second number is 0.
- If it is 0, display **"Cannot divide by zero"**.
- Use try-except to handle letters or invalid number input.
- If the user enters letters, display **"Please enter numbers only"** instead of crashing.

Sample Input

Enter first number: 40

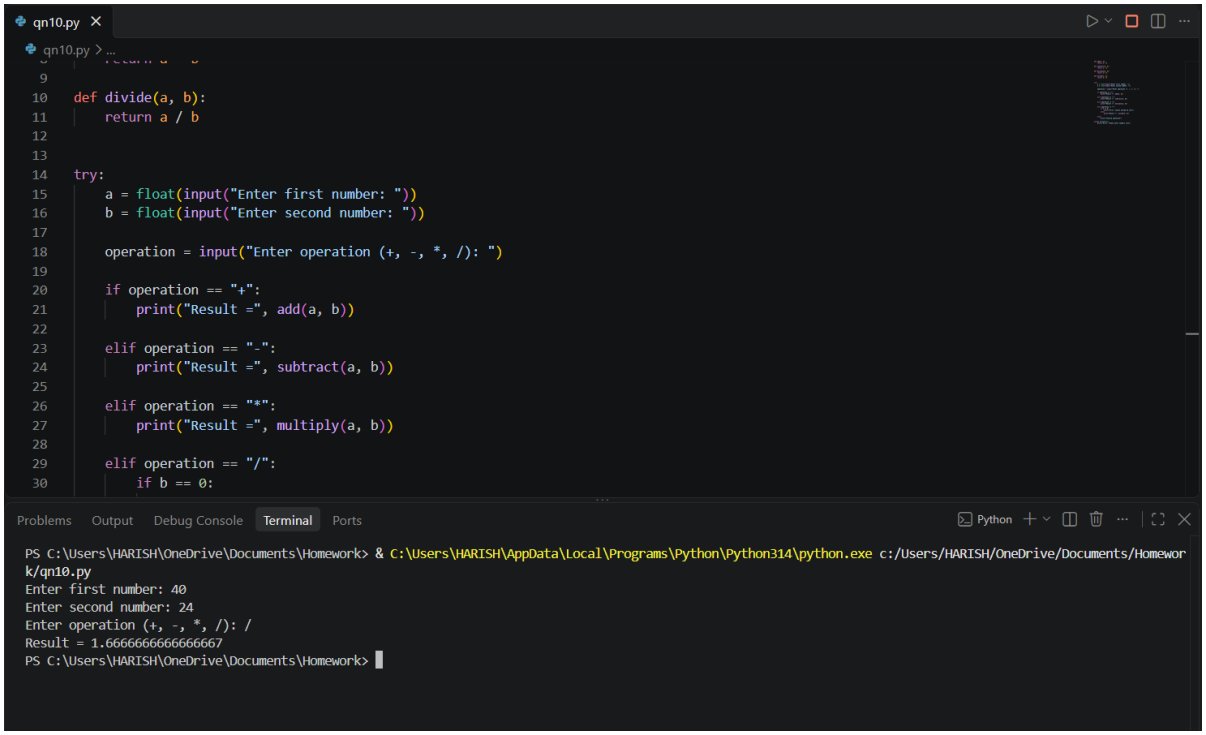
Enter second number: 24

Enter operation (+, -, *, /): /

Sample Output

Result = 1.6666666666666667

Result


qn10.py
def divide(a, b):
return a / b
try:
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
operation = input("Enter operation (+, -, *, /): ")
if operation == "+":
print("Result =", add(a, b))
elif operation == "-":
print("Result =", subtract(a, b))
elif operation == "*":
print("Result =", multiply(a, b))
elif operation == "/":
if b == 0:
Problems
Output
Debug Console
Terminal
Ports
Python
PS C:\Users\HARISH\OneDrive\Documents\Homework> & C:\Users\HARISH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/HARISH/OneDrive/Documents/Homework/qn10.py
Enter first number: 40
Enter second number: 24
Enter operation (+, -, *, /): /
Result = 1.6666666666666667
PS C:\Users\HARISH\OneDrive\Documents\Homework>